

Causal Probabilistic Transformer: A Probabilistic Dependency Model for Autoregressive Language Modelling

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Abstract

The probabilistic transformer of Wu and Tu (2023) models contextual word representation as posterior inference in a conditional random field over dependency structure, and its mean-field inference procedure turns out to resemble a transformer encoder. That model is an encoder only. In this work we derive its autoregressive counterpart. We show that causality, a single global energy function and contextuality cannot hold simultaneously, and resolve the conflict with a directed chain of conditional random fields in which the prefix enters each step as a frozen condition. We further obtain a language-model head without adding any parameter outside the factor graph, by promoting the word variable from observed to latent; tied input and output embeddings then follow as a consequence rather than a design choice. Because the per-position subgraph is a tree, the vocabulary readout is available in closed form. Experiments on [TODO: corpora] compare our model against a standard causal transformer and against a looped transformer, the latter isolating the effect of syntactic structure from that of weight sharing. [TODO: one sentence of result] We hope that this work helps extend the probabilistic account of contextual representation to generation.

1 Introduction

Transformers are usually designed rather than derived. Wu and Tu (2023) take the opposite route: they define a conditional random field over the discrete latent labels of the words in a sentence and over the dependency arcs between them, perform approximate inference with mean field variational inference, and observe that the resulting computation graph closely resembles a transformer encoder. Self-attention appears there not as a component that was chosen, but as the posterior distribution over syntactic dependency heads. The model is

not an attempt to improve on transformers, but an account of what one of them might be doing.

That account stops at the encoder. It is trained with masked language modelling, whereas transformers are now predominantly used autoregressively. Wu and Tu (2023) name a decoder as an open extension and do not pursue it. We ask whether the probabilistic account survives the move to generation, or whether it was a property of the encoder all along.

Masking the dependency factors so that a head may only point backwards is not enough. Under a single global energy the marginals at two positions share a normalising constant, so a factor coupling them reaches the earlier one whenever it reaches the later one; masking changes which configurations score highly, not that the survivors are normalised jointly. Causality, a single global energy, and genuine contextual dependence cannot hold together, and the first and third are what the model is for. We therefore replace the single field with a directed chain of conditional random fields, one per position, in which the prefix posteriors enter as a frozen condition, so that the anti-causal contribution is never formed rather than formed and deleted.

A second problem has no counterpart in the encoder: a model that predicts words needs a distribution over the vocabulary. Projecting the label posterior through a matrix into vocabulary space would add a component corresponding to no factor, so the prediction would leave the model without traversing the graph. No new component is needed. The word is already a variable of the graph, merely observed, and word-label compatibility is already a factor of it; unclamping the variable turns that factor into a language model head. Properties that are usually stipulated then follow: the input and output word embeddings are the same matrix, since one factor mediates both directions; the output bias is the unary term on the word variable; and encoding and decoding are two conditioning patterns of one

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model rather than two models.

Both results are refusals, and they refuse for the same reason: the factor graph is the model, and it constrains what may be added to it. There is also a dividend. Conditioned on its prefix, the subgraph at a single position is a star centred on the label variable, hence a tree, so inference there is exact and the readout is a mixture of softmaxes in closed form, computed along the sequence by a single causal scan that leaves the layer-parallel schedule intact.

Empirically we ask whether syntactic structure pays for itself. A comparison against a standard decoder cannot settle this, because our model differs from one in two ways at once: it reuses a single parameter set across inference iterations, and it has a syntactic factor graph. A looped transformer, in which one block is applied repeatedly with shared weights, has the first without the second, so the comparison against it is the one that isolates structure. Following Wu and Tu (2023), our aim is not to compete with transformers on perplexity; a gap is expected, since all predictive information passes through the label variables and the readout inherits their rank. **[TODO: one sentence of measured result.]** We hope this work extends the probabilistic account of contextual representation from analysis to generation.

2 Causal Probabilistic Transformers

We first recall the model of Wu and Tu (2023) only as far as we need it (Section 2.1), then state the obstruction that prevents a direct causal version (Section 2.2), then give the model (Section 2.3), its output mechanism (Section 2.4), its inference procedure (Section 2.5), and finally some variants.

2.1 Background

[verify against PDF: Wu & Tu §2.1–§2.2 — variable set, the two potential functions, MFVI]
[TODO: compress to about half a column. Only what Sections 2.2 and 2.4 actually use: the label variables Z_i over d labels; the head variables $H_i^{(c)}$ over candidate heads, one set per channel; the unary factor $\phi_u(Z_i) = \exp(S_{w_i, Z_i})$ with $S \in \mathbb{R}^{|\mathcal{V}| \times d}$; the ternary factor over $(H_i^{(c)}, Z_i, Z_j)$ with score matrix $T^{(c)}$; and that MFVI unrolled for T iterations has the shape of a T -layer transformer encoder with shared weights. Note explicitly that the word w_i enters only through S — Section 2.4 turns on that fact.]

2.2 Why a causal mask is not enough

[TODO: about half a column. Wu & Tu’s paper contains no theorem environments; keep this to one theorem statement in prose-adjacent form and push the proof to Section C.]

The obvious route to a causal model is to mask the ternary factors so that $H_i^{(c)}$ may only point to positions $j \leq i$. This does not give a causal model. Informally, three properties that one would want cannot hold together:

- **Causality:** the posterior at position i depends only on $w_{1:i}$;
- **A single global energy:** one energy function over the sentence, normalised once;
- **Contextuality:** the representation at i genuinely depends on its prefix.

Theorem 1 (Trilemma). **[TODO: formal statement]** *No model satisfies all three simultaneously.*

The mechanism is the partition function: a factor coupling positions $i < j$ contributes to the marginal at i whenever it contributes to the marginal at j , and forbidding the first while keeping the second is inconsistent with a single normalisation. Masking the factors changes which configurations have non-zero score; it does not stop the surviving ones from being normalised jointly. The proof is in Section C.

Causality and contextuality are what the model is for, so the single global energy is what we give up.

Remark 2 (Never formed, not deleted). Building the symmetric model and then removing the anti-causal contribution gives a different object from building a model in which that contribution never exists. This distinction is what Section 2.3 implements and what Section 4.3 tests.

2.3 A directed chain of conditional CRFs

[verify against PDF: Part II of the main document — the factorisation and the conditioning statement]

Instead of one CRF over the sentence, we define a chain of conditional CRFs, one per position. The CRF at position t ranges over $(Z_t, H_t^{(1..h)}, W_t)$ and is conditioned on the prefix posteriors $\{Q_{Z_s}\}_{s < t}$, which enter as fixed numbers:

[TODO: the factorisation] (1)

[TODO: Figure 1. Two panels of the *same* factor graph: (a) encoder, W_i shaded (observed); (b) decoder, W_i unshaded (latent), with the prefix boxed as a frozen condition. Same factors, same edges, different shading. This single figure carries the claim that encoding and decoding are two conditioning patterns of one model, and it is the analogue of Wu & Tu’s Figure 1, which is also introduced at exactly this point.]

Figure 1: The factor graph of the causal probabilistic transformer for $n = 3$.

The prefix is a frozen condition. Each Q_{Z_s} , $s < t$, is data at step t : no message travels from t back to s . The anti-causal term of Theorem 1 is therefore never formed, and causality holds by construction rather than by masking.

Frozen forward, not frozen backward. *Frozen* means constant with respect to step t ’s inference problem, not detached in autodiff. Training gradients flow backwards through the whole prefix computation, exactly as they do through cached activations in a causal transformer; this is how S is trained in both of its roles (Section 2.4). [note: say this explicitly in the paper. It is the single easiest thing for a reader to get backwards, and getting it backwards makes the model look either non-causal or untrainable.]

2.4 The output mechanism

This is the part of the construction we most want checked, and we present it as such.

The design we rejected. The obvious way to obtain next-token logits is to project Q_{Z_t} through a fresh matrix in $\mathbb{R}^{d \times |\mathcal{V}|}$. Such a matrix corresponds to no factor in the graph: the output would not traverse the factor graph at all, and the model would be a CRF with a neural head attached. We report this because rejecting it is what forced the mechanism below.

Unclamping the word variable. The word variable is already in the graph and S is already a binary factor; in the encoder it is evaluated with its word argument clamped to the observed token. The output mechanism is to stop clamping it:

[TODO: $\mu_t(a) \propto \dots$ in the notation of causal_pt_terminator and fix (2)]

No factor is added and no parameter is created. Next-token prediction is the posterior over a variable the model already had.

Consequences. Three properties follow, and we stress that they are consequences rather than design choices:

1. **Tying is forced.** One factor S mediates both directions, so input and output word embeddings are the same matrix. Untying is not a hyperparameter; it is a different model.
2. **The output bias is the word unary.** [TODO: state]
3. **One model, two conditioning patterns.** The encoder clamps W_i ; the decoder unclamps it and directs the chain. Same parameters, same factors (Figure 1).

2.5 Inference

[note: Resolved 2026-08-05: §17.2 proposes MFVI as the mainline but §23.3 supersedes it. The exact readout is the model; the mean-field readout is the ablation of Section 4.3. Do not present them as two equal options.]

[verify against PDF: Part III §16–§17.1 — update equations and schedule]

$$Q_{Z_t} \propto [\text{TODO:}] \quad (3)$$

$$Q_{H_t^{(c)}} \propto [\text{TODO:}] \quad (4)$$

Exact readout. The subgraph at position t is a star centred on Z_t , hence a tree, so sum-product is exact: no approximation and no iteration are involved.

Proposition 3. [verify against PDF: §23.3 — confirm the form and the seeding by $r_a^{(c)}$]

$$\log \mu_t(a) = \sum_{c=1}^h \text{LSE}_{j \in \mathcal{D}_t} B_{j,a}^{(c)}, \quad (5)$$

seeded with the root key $r_a^{(c)}$.

Along the sequence $\mathcal{D}_t$ grows by one position per step, so Equation (5) is a causal prefix log-sum-exp: one logcumsumexp scan, $O(ndh)$, fully parallel over positions. The exact readout therefore costs neither the layer-parallel schedule nor an inner loop — with it there is no query stream and no predictive iteration at all. Two costs are real and we state them: the $|\mathcal{V}| \times d$ readout is a log-sum-exp rather than a matmul (same FLOPs,

worse hardware constants, chunked over the vocabulary with a fused cross-entropy), and LSE-pooling gradients are untested at language-model scale.

Every step is valid inference. [verify against PDF: Part III §18 Check 3, as restated in §23.3]

Mean field where exactness is impossible (the chain), exact sum-product where the subgraph is a tree (the slot).

2.6 Extensions and variants

Distance. [TODO: the causal half of the clipped ternary potential: only $i - j > 0$ exists, so there are $\gamma + 1$ buckets rather than $2\gamma + 1$.]

Update schedule. [TODO: parallel (a depth- T shared causal transformer) vs. serial (τ inner rounds per slot); say which produced the reported numbers.]

Global variables. [TODO: the Appendix B.3 globals G_t , proposed by Wu and Tu (2023) as an in-graph substitute for the missing feed-forward structure and never tested there. We treat them as a measured variable (Section 4.3), not an assumption.]

Further variants are in Section B.

3 Comparison with Causal Transformers

Wu and Tu (2023) show that MFVI in the encoder model reproduces the computation graph of a transformer encoder, with the posterior over dependency heads playing the role of self-attention. We ask the same question of the causal model, and the answer is the reason the experiments in Section 4 take the shape they do.

3.1 Tensor form

[TODO: re-derive the update equations of Section 2.5 in tensor form, following the structure of Wu & Tu §3.1 so the two papers can be read side by side. Their $Q_z^{(t)}$, $Q_{h,c}^{(t)}$, $\mathcal{F}$, $\mathcal{G}$ notation should be reused verbatim wherever the quantity is the same one — deviating from it without reason costs the reader the comparison.]

3.2 Single-channel update vs. masked self-attention

[TODO: the causal analogue of their §3.2. The claim to establish: the single-channel update under the causal chain reduces to scaled dot-product attention with a causal mask, and the correspondence $Q_c \leftrightarrow W^Q$, $K_c = V_c \leftrightarrow W^K =$

Model	Weight sharing	Syntactic structure
Causal transformer	no	no
Looped transformer	yes	no
Causal PT	yes	yes

Table 1: The three models of Section 4. Looped vs. transformer isolates weight sharing; causal PT vs. Looped isolates structure, which is the comparison this paper is about.

W^V carries over. State precisely where it stops holding — that is more interesting than where it does.]

3.3 The readout vs. a language-model head

[TODO: no counterpart in Wu & Tu, because the encoder has no LM head. Show that Equation (5) is a mixture of h softmaxes over a d -dimensional space, compare with the ordinary $\mathbb{R}^{d \times |\mathcal{V}|}$ head, and connect to Yang et al. (2018): the rank- d bottleneck is real, but a mixture of softmaxes is not rank-limited in the way a single softmax is. Whether that helps is measured in Section 4.3.]

3.4 Full model comparison

[TODO: prose. The differences to draw out, following their §3.4:]

- **No feed-forward structure.** Inherited from the encoder model, and the reason the global variables of Section 2.6 are worth measuring.
- **No residual connections or layer norm.** The unary scores are re-added at each iteration instead.
- **Parameters shared across iterations.** This is the property the Looped transformer (Dehghani et al., 2019; Giannou et al., 2023) also has, and it is why a comparison against a standard decoder alone cannot attribute any difference to structure.
- **The output is a posterior, not a projection.** No parameter exists outside the factor graph (Section 2.4).

Table 1 is what the experimental design follows from: the causal PT differs from a standard decoder in two ways at once, so a two-model comparison cannot attribute a difference to either. [note:

[TODO: Figure 2, full width. Three computation graphs side by side — (a) GPT-style decoder, (b) Looped decoder, (c) causal probabilistic transformer — in the visual style of Wu & Tu’s Figure 3. This figure is what makes Table 1 unnecessary to argue for in words.]

Figure 2: Computation graphs for a causal transformer, a Looped transformer, and a causal probabilistic transformer.

putting this table in Section 3 rather than Section 4 is deliberate — it makes the Looped baseline a consequence of the model comparison instead of an experimental choice that a reviewer could read as cherry-picked.]

4 Experiments

[note: No number appears here without a row in the run log of the corresponding experiments/*/EXPERIMENT_STATUS.md: date, commit, config, seed, metric, wall-clock.]

4.1 Tasks and Datasets

[TODO: brief, with the detail in Section D — this is how Wu & Tu handle it.]

Language modelling. [TODO: PTB (Marcus et al., 1993) and/or WikiText-2 (Merity et al., 2017); perplexity on the held-out set.]

We deliberately do not use WikiText-103. Wu and Tu (2023) report that the encoder model significantly underperforms transformers beyond roughly 10^5 sentences, suspected to be the absence of a feed-forward structure; running there would test a known failure mode of the encoder rather than the causal extension. [note: state this before a reviewer asks why the corpus is small — their own paper makes the same point in its Discussion]

4.2 Settings

[TODO: parameter budget 20–50M matched across models; identical tokenizer, vocabulary and context length; one shared training loop; Adam with identical hyperparameters and schedule; k seeds, mean $\pm$ std. Wu & Tu report mean and std over 5 random runs — match that.]

Matching budgets. We report parameter count and wall-clock, since the causal PT shares parameters across iterations and equal parameter count therefore does not mean equal compute.

The embedding split. We report embedding and non-embedding parameters separately, which matters more here than in a usual comparison. With

tying forced (Section 2.4) the causal PT’s parameters sit almost entirely in $S \in \mathbb{R}^{|\mathcal{V}| \times d}$; the arc scores $T^{(c)}$, the word unary b and the root key $r^{(c)}$ are negligible beside it. At a matched total the causal PT is close to an embedding table plus a little and a transformer baseline is not, and a single total hides that completely. [TODO: state the matching rule chosen — total parameters, non-embedding parameters, or both with vocabulary fixed. Any is defensible; leaving it unstated is what gets attacked first.]

Loop depth. The loop depth of the Looped baseline matches the number of MFVI iterations T , so effective depth is comparable as well as parameter count. We report a sweep over T rather than a single value, so the comparison does not rest on one lucky setting.

4.3 Results

Correctness of the implementation. [TODO: two sentences plus Section E. Only the two checks that test the derivation rather than the code belong in the main text: the mean-field free energy is non-increasing across iterations on a toy graph, and Equation (5) agrees with brute-force enumeration over a tiny vocabulary to [TODO: 1e-N]. Causality, tying, normalisation and the single-batch overfit are software checks — list them in the appendix.]

Language modelling. [TODO: read Table 2. Two readings, in this order: (i) the causal PT trains and lands within a modest gap of the transformer — necessary but not the point; (ii) the causal PT vs. Looped gap is the paper’s claim, because Looped has the weight sharing without the structure (Table 1). If the causal PT is worse than Looped across the T sweep, the central claim fails and we say so here.]

Ablations. [TODO: (i) global variables G_t on/off — a measured variable, not an assumption, and the delta is itself a result since Wu and Tu (2023) propose them but never test them; (ii) exact vs. mean-field readout, below.]

Dataset	Metric	Params (emb. / non-emb.)	Transformer	Looped	Causal PT
PTB	Perplexity	[TODO:]	[TODO:]	[TODO:]	[TODO:]
WikiText-2	Perplexity	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 2: Main results, mean $\pm$ std over [TODO: k] random runs. The comparison this paper is about is the last two columns.

Exact vs. mean-field readout. [TODO: Two variants. (a) Evaluation-time swap: train with the mean-field readout, substitute Equation (5) at evaluation on the same parameters — isolates the approximation error, nearly free. (b) Train with each — measures the end-to-end effect, one extra full run. Neither outcome is a win or a loss: a small gap says mean field is adequate and cheap, a large one says the tree structure should be exploited. Also check whether the mixture-of-softmaxes form relieves the rank- d bottleneck of Section 3.3.]

5 Related Work

[TODO: one paragraph, in this order:]

- **Syntax in transformers.** [TODO: the line Wu and Tu (2023) cite — supervised attention to syntactic governors, constituency/dependency injection, dependency-constrained attention. Position this work as the same deviation they claim: we start from probabilistic modelling and arrive at a transformer, rather than adding syntax to one.]
- **Weight sharing and looped models.** [TODO: Dehghani et al. (2019), Giannou et al. (2023), Yang et al. (2024). This citation is load-bearing: it establishes that the Looped baseline of Section 4 is a recognised model and not a strawman constructed for this paper.]
- **Structured autoregressive models.** [TODO: what exists for causal LMs with latent structure, and why none of it is a causal PT.]
- **The softmax bottleneck.** [TODO: Yang et al. (2018); connect to Section 3.3.]

6 Discussion

What this paper does and does not claim. As in Wu and Tu (2023), our goal is not to propose

a model that competes with transformers on perplexity. It is to establish that the probabilistic-dependency account of contextual representation extends to autoregressive generation, and to measure what its structure contributes. [TODO: one sentence stating what the numbers actually showed, written after they exist and not before.]

The rank- d bottleneck. [TODO: All predictive information reaches the vocabulary through Z_t , whose support has size d . This is an honest cost of the construction rather than a defect of the implementation, and it predicts a gap to a standard decoder that no amount of tuning removes. Say by how much, and relate to the measurement in Section 4.3.]

The missing feed-forward structure. [TODO: inherited from the encoder model, and the leading suspect in its degradation on larger corpora. Report what the global-variable ablation of Section 4.3 found — this is the first empirical test of the Appendix B.3 proposal of Wu and Tu (2023), and it is worth saying so explicitly whichever way it came out.]

Extensions. [TODO: keep short. Candidates: continuous or vector-valued Z ; dependency-label variables paired with each H^0 ; whether the exact readout suggests anything transferable back to ordinary decoders — which is the direction Wu and Tu (2023) name as the point of the whole programme.]

7 Conclusion

[TODO: one paragraph, written last and written to the measured result. If the causal PT matches Looped, the claim is that syntactic structure carries information weight sharing does not. If it does not, the contribution is the construction plus a negative empirical finding, which is still a paper — do not reframe the numbers to rescue the stronger claim.]

Limitations

[TODO: three or four concrete items. Candidates, each a consequence of the construction rather than something more tuning would fix:]

- **The rank- d bottleneck.** All predictive information reaches the vocabulary through the label variables (Section 3.3).
- **Scale.** Evaluated only on small corpora, for the reason given in Section 4.1; behaviour beyond that regime is untested here, and the encoder model is reported to degrade there.
- **No single sequence-level joint.** Giving up the global energy (Section 2.2) means quantities defined by that joint are unavailable.
- **Speed.** **[TODO: measure. The vocabulary readout is a log-sum-exp rather than a matmul: same FLOPs, worse constants. Wu & Tu report their encoder as about 3 times slower than a transformer despite having far fewer parameters — report the analogous number here rather than leaving the reader to assume it is small.]**
- **[TODO: at least one item that the runs exposed rather than the construction predicted. A limitations section made entirely of things known in advance is less credible than one that admits a surprise.]**

References

- Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Łukasz Kaiser. 2019. Universal transformers. In *International Conference on Learning Representations*.
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A Derivations

A.1 Full mean-field updates

[TODO: the complete updates from Part III §16, with the steps Section 2.5 compresses.]

A.2 Derivation of the exact readout

[TODO: sum-product on the star subgraph, giving Equation (5); and the logcumsumexp form along the sequence.]

A.3 Message weights

[TODO: $\lambda_Z, \lambda_H, \lambda_W, \lambda_G$ — follow Wu & Tu §A.5, and state where the causal model forces a different choice from theirs.]

A.4 Complexity

[TODO: per-token cost of the content-stream solve, the $O(dh)$ cache update and the vocabulary readout; and the chunked fused cross-entropy used in the implementation.]

B Additional Variants

B.1 Global variables

[TODO: the G_t construction, following Appendix B.3 of Wu and Tu (2023), and what changes in the causal setting. The measurement is in Section 4.3; this is the model description.]

B.2 Serial update schedule

[TODO: τ inner rounds per observed slot; when it differs from the parallel schedule and by how much.]

B.3 Root node

[TODO: whether the dummy root of Wu & Tu §2.3.5 survives in a causal model, and what the root key $r^{(c)}$ of Equation (5) is in relation to it.]

C Proof of Theorem 1

[TODO: the full statement and proof, from Part I of the main document. Section 2.2 carries only the statement and the mechanism in one paragraph, which is what the format allows; everything else is here.]

[TODO: state the three properties formally as they are compressed to bullets in the main text: causality, a single global energy, contextuality. Then the incompatibility argument.]

D Datasets

[TODO: corpus statistics, preprocessing, tokenisation, vocabulary construction, train/dev/test splits, and the exact sequence length. Enough for someone to reproduce the numbers in Table 2 without reading the code.]

E Implementation Validation

[TODO: the full checklist, with pass/fail and the numbers. Section 4.3 reports only the two that test the equations; these are the rest.]

- **Causality.** Changing the token at position t leaves the logits at every position $< t$ bitwise unchanged (CPU, fixed seed — on GPU, non-deterministic reductions can perturb the last bit of correct code).
- **Frozen prefix.** Prefix quantities are kept as attached leaves with `requires_grad=True`; after backpropagating the loss at step t , their gradient is exactly zero for $j \geq t$ and non-zero for $j < t$. Testing this on a detached tensor would pass unconditionally and is therefore no test at all.
- **Tying.** Input and output embeddings are the same tensor object, not merely equal.
- **Normalisation.** All posteriors sum to one along their variable axis at every iteration.
- **Free energy.** The mean-field free energy is non-increasing across iterations on a toy graph. [TODO: figure] [note: this is the check that catches a typo in the update equations; shapes, normalisation and causality are all blind to one.]
- **Exact readout vs. brute force.** On one slot the graph is a tree, so Equation (5) must agree

with explicit enumeration over a tiny vocabulary. [TODO: agreement to 1e-N] This validates the factorisation itself.

- **Single-batch overfit.** Loss on one batch falls to near zero.

F Worked Example

[TODO: reproduce causal_pt_output_note.pdf §5 — $\mathcal{V} = \{the, cat, sat, mat\}$, $d = 2$, one channel, both readout modes, with printed posteriors and logits at every step.]

[note: take the numbers from the test-suite output rather than retyping them, so the appendix and the tests cannot drift apart, and say in the caption that they agree with the independently derived reference.]

[TODO: if space allows, add the analogue of Wu & Tu’s Appendix F proper: a case study of the dependency structures the causal model infers on real text. Their finding was that the induced structures are only partially consistent with human intuition — reporting the same honestly is better than omitting the section.]